

Presidential Speech Sentiment Analysis

227 U.S. State of the Union Addresses

Analysis of Results

Sentiment Count

Positive and negative word counts with net sentiment score from the first speech.

negative	positive	sentiment
59	89	30

Interpretation: Adams' 1797 address contained 59 negative and 89 positive words, yielding a net sentiment score of 30. The positive word count outweighing the negative indicates an overall positive tone in the speech.

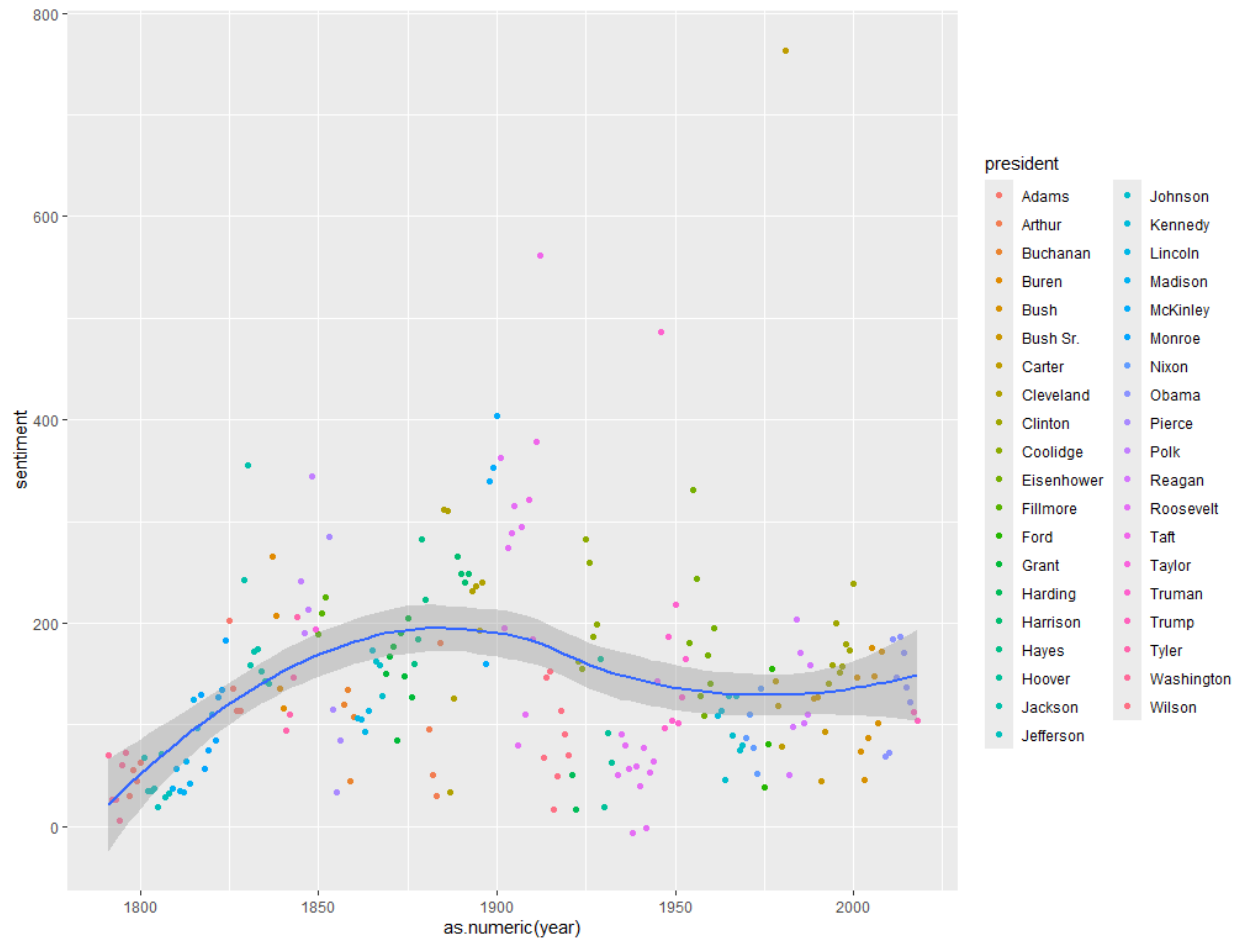
Summary Statistics

Descriptive statistics for sentiment variables across all 227 addresses.

variable	min	1st qu.	median	mean	3rd qu.	max
negative	Min.: 20.0	1st Qu.: 101.0	Median: 138.0	Mean: 177.5	3rd Qu.: 209.5	Max.: 820.0
positive	Min.: 57.0	1st Qu.: 180.5	Median: 282.0	Mean: 321.3	3rd Qu.: 386.5	Max.: 1458.0
sentiment	Min.: -7.0	1st Qu.: 76.0	Median: 128.0	Mean: 143.8	3rd Qu.: 184.0	Max.: 763.0
year	Min.: 1791	1st Qu.: 1848	Median: 1904	Mean: 1904	3rd Qu.: 1962	Max.: 2018

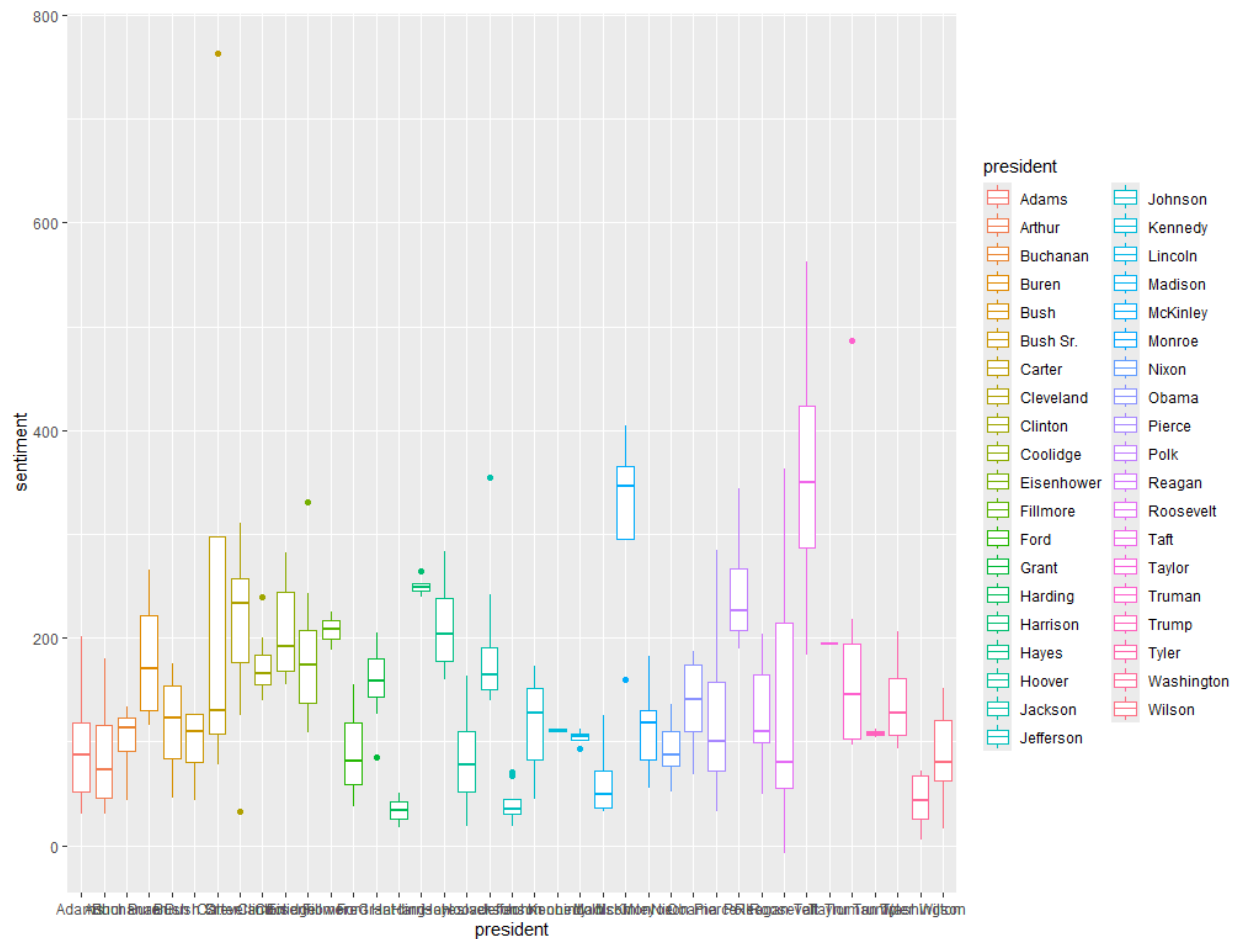
Interpretation: Across all 227 addresses, positive word counts consistently exceeded negative counts, with a mean of 321.3 positive words compared to 177.5 negative words per speech. The median net sentiment score of 128 and mean of 143.8 suggest that most speeches lean positive in tone. The wide range in sentiment (-7 to 763) indicates considerable variability across addresses, with a small number of speeches driving the upper extreme.

Sentiment Scores in U.S. State of the Union Addresses Over Time (1791–2018)



Interpretation: Sentiment scores rose through the 1800s before peaking around the turn of the 20th century, followed by a period of high variability in the early 1900s and a notable dip around mid-century. Scores stabilized moderately in the modern era. The smoothed trend line suggests that presidential tone has fluctuated across history rather than following a consistent upward or downward trajectory.

Sentiment Score Distribution by President Across U.S. State of the Union Addresses



Interpretation: Sentiment scores vary considerably both within and across presidents. Several presidents, including Reagan and Roosevelt, display wide interquartile ranges and notable outliers, while others maintain more consistent tone across their addresses. The degree of within-president variability suggests that individual speech context may matter more than a president's overall rhetorical style.

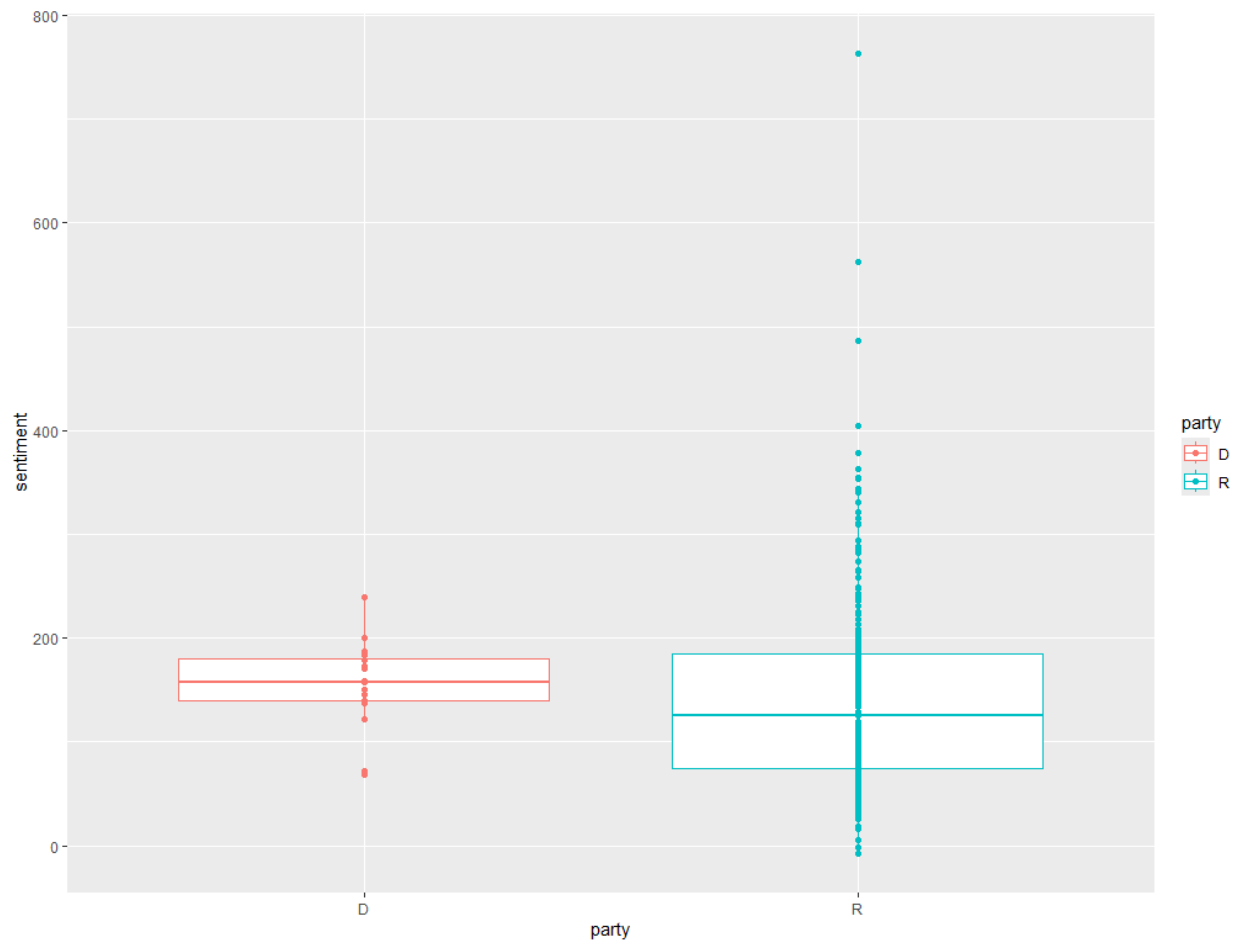
Party Sentiment Comparison: Welch Two Sample t-test

Testing whether mean sentiment scores differ significantly between Democratic and Republican presidents.

statistic	value
Test	Welch Two Sample t-test
Groups	Democrats vs. Republicans
t-statistic	0.96626
Degrees of Freedom	29.756
p-value	0.3417
95% CI	-13.87927 to 38.79041
Mean (Democrats)	155.3750
Mean (Republicans)	142.9194
Conclusion	Fail to reject H_0 – no statistically significant difference

Interpretation: Democratic presidents averaged a slightly higher sentiment score (155.4) compared to Republican presidents (142.9). However, with $t = 0.97$, $df = 29.76$, and $p = 0.3417$, the difference is not statistically significant at the $\alpha = 0.05$ level. The 95% confidence interval (-13.88 to 38.79) spans zero, further supporting the conclusion that party affiliation does not significantly predict sentiment score.

Sentiment Score Distribution by Political Party in U.S. State of the Union Addresses (1791–2018)



Interpretation: The sentiment distributions for Democratic and Republican presidents overlap substantially, consistent with the t-test result. While Democrats show a marginally higher median and Republicans display slightly greater spread, neither pattern is strong enough to suggest that party affiliation meaningfully drives differences in the emotional tone of State of the Union addresses.

References

Tatman, R. (2017). *Tutorial: Sentiment analysis in R* [Kaggle notebook]. Kaggle.

<https://www.kaggle.com/code/rtatman/tutorial-sentiment-analysis-in-r/notebook>

Attachments

R Outputs

```
1 # load in the libraries we'll need
2 library(tidyverse)
3 library(tidytext)
4 library(glue)
5 library(stringr)
6 library(dplyr)
7 library(tidyr)
8
9 # get a list of the files in the input directory
10 input_folder <- "[REDACTED] State of the Union Corpus (1790 - 2018)/sotu"
11 files <- list.files(input_folder)
12
```

```

14
15 # stick together the path to the file & 1st file name
16 fileName <- glue("{input_folder}/{files[1]}")
17 # get rid of any sneaky trailing spaces
18 fileName <- trimws(fileName)
19
20 # read in the new file
21 fileText <- glue(read_file(fileName))
22 # remove any dollar signs (they're special characters in R)
23 fileText <- gsub("\\$", "", fileText)
24
25
26 # tokenize
27 # got a data_frame() warning so fixed the code
28 tokens <- tibble(text = fileText) %>% unnest_tokens(word, text)
29 sentiments <- tibble()
30
31 # get the sentiment from the first text:
32 tokens %>%
33   inner_join(get_sentiments("bing")) %>% # pull out only sentiment words
34   count(sentiment) %>% # count the # of positive & negative words
35   spread(sentiment, n, fill = 0) %>% # made data wide rather than narrow
36   mutate(sentiment = positive - negative) # # of positive words - # of negative words
37
38 # write a function that takes the name of a file and returns the # of positive
39 # sentiment words, negative sentiment words, the difference & the normalized difference
40 GetSentiment <- function(file) {
41   fileName <- file.path(input_folder, file)
42
43   fileText <- read_file(fileName) %>%
44     gsub("\\$", "", .) %>%
45     gsub("[^[:alnum:]]\\s+", " ", .) %>%
46     gsub("\\s+", " ", .)
47
48   tokens <- tibble(text = fileText) %>%
49     unnest_tokens(word, text) %>%
50     mutate(word = str_to_lower(word))
51
52   sentiment_counts <- tokens %>%
53     inner_join(get_sentiments("bing"), by = "word", relationship = "many-to-many") %>%
54     count(sentiment) %>%
55     pivot_wider(names_from = sentiment, values_from = n, values_fill = 0)
56
57   if (!"positive" %in% colnames(sentiment_counts)) sentiment_counts$positive <- 0
58   if (!"negative" %in% colnames(sentiment_counts)) sentiment_counts$negative <- 0
59
60   sentiment_counts %>%
61     mutate(
62       sentiment = positive - negative,
63       file = file,
64       year = as.numeric(str_extract(file, "\\d{4}")),
65       president = str_match(file, "(.*?)"[2]
66     )
67 }
68
69 # test: should return
70 # negative positive sentiment file year president
71 # 117 240 123 Bush_1989.txt 1989 Bush
72 # Note** it's returning Adams_1797 because it is the first .txt in folder
73 GetSentiment(files[1])
74

```

```

75 # file to put our output in
76 sentiments <- data_frame()
77
78 # get the sentiments for each file in our dataset
79 for(i in files){
80   sentiments <- rbind(sentiments, GetSentiment(i))
81 }
82
83 # disambiguate Bush Sr. and George W. Bush
84 # correct president in applicable rows
85 bushsr <- sentiments %>%
86   filter(president == "Bush") %>% # get rows where the president is named "Bush"...
87   filter(year < 2000) %>% # ...and the year is before 200
88   mutate(president = "Bush Sr.") # and change "Bush" to "Bush Sr."
89
90 # remove incorrect rows
91 sentiments <- anti_join(sentiments, sentiments[president == "Bush" & sentiments$year < 2000, ])
92
93 # add corrected rows to data_frame
94 sentiments <- full_join(sentiments, bushsr)
95
96 # summarize the sentiment measures
97 summary(sentiments)
98
99 # plot of sentiment over time & automatically choose a method to model the change
00 ggplot(sentiments, aes(x = as.numeric(year), y = sentiment)) +
01   geom_point(aes(color = president)) + # add points to our plot, color-coded by president
02   geom_smooth(method = "auto") # pick a method & fit a model
03
04 # plot of sentiment by president
05 ggplot(sentiments, aes(x = president, y = sentiment, color = president)) +
06   geom_boxplot() # draw a boxplot for each president
07
08 # is the difference between parties significant?
09 # get democratic presidents & add party affiliation
10 democrats <- sentiments %>%
11   filter(president %in% c("Clinton", "Obama")) %>%
12   mutate(party = "D")
13
14 # get democratic presidents & party add affiliation
15 republicans <- sentiments %>%
16   filter(!president %in% c("Clinton", "Obama")) %>%
17   mutate(party = "R")
18
19 # join both
20 byParty <- full_join(democrats, republicans)
21
22 # the difference between the parties is significant
23 t.test(democrats$sentiment, republicans$sentiment)
24
25 # plot sentiment by party
26 ggplot(byParty, aes(x = party, y = sentiment, color = party)) + geom_boxplot() + geom_point()
27
28

```